

A simulator for bidirectional ant traffic

Model, parameters and methods

Technical companion to the `simulateur/` directory

September 15, 2026

Abstract

This document records the model as actually implemented in the `simulateur/` directory, the provenance of each of its parameters, and the protocols used to measure the observables. It follows on from the hand-off report *An individual-based model for bidirectional ant traffic on a narrow bridge*, whose microscopic model and unit conventions it keeps; it departs from it on five points, all listed in §6 together with the measurement that motivates each. The guiding principle is that every number should be traceable: for each parameter we state whether it is *measured* on the experimental trajectories, *calibrated* against a macroscopic observable, or *inherited* from the reference model without any new constraint.

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1 Scope

The simulator lives in `simulateur/`, independently of the reference code, which it does not modify. It implements the same microscopic model — decoupled radial/angular dynamics, interaction repertoire, thigmotaxis — with an architecture rebuilt to make parameter sweeps and reproducible measurements possible.

Internal units are those of the hand-off report: **1 length unit = 1 mm**, **1 time unit = 1 s**, ant of radius $R = 1$. Conversions to the units of Poissonnier et al. (2019) are $k [\text{cm}^{-2}] = \rho [\text{mm}^{-2}] \times 100$ and $q [\text{cm}^{-1}\text{s}^{-1}] = 10 \rho U$.

2 The model

2.1 Variables and dynamics

Each agent i carries a position $X_i \in \mathbb{R}^2$, a *scalar* speed $u_i \geq 0$, a heading θ_i and a fixed group $g_i \in \{-1, +1\}$ (+1 towards the food, -1 towards the nest). The vector velocity is $V_i = u_i \hat{\Theta}_i$ with $\hat{\Theta}_i = (\cos \theta_i, \sin \theta_i)$. Positions evolve through $\dot{X}_i = u_i \hat{\Theta}_i$, and the two velocity components obey two separate relaxation equations:

$$\frac{du_i}{dt} = (\nu_c + \nu_b) \left(\frac{\nu_c}{\nu_c + \nu_b} \xi_i - u_i \right), \quad (1)$$

$$\frac{d\theta_i}{dt} = \lambda \sin(\omega_i - \theta_i), \quad \omega_i = \arg F_i. \quad (2)$$

Equation (1) is the core of the model: braking enters as a *rate* of relaxation, not as a subtracted force. Under a mono-kinetic closure where $\nu_b \simeq \bar{\beta}k$, the equilibrium speed is $U(k) = \xi \nu_c / (\nu_c + \bar{\beta}k)$ and the flow $q = kU$ *saturates* instead of collapsing — this is the mechanistic origin of the plateau.

2.2 Angular dynamics: three modes

Written as it stands, the hand-off report’s equation ($\lambda = |F_i|$) makes the norm of the field play two roles at once: it sets the target direction *and* the rate at which the agent turns towards it. A strong repulsion term therefore makes the turn nearly instantaneous, which forces a cap on $|\dot{\theta}|$ — a cap that then mixes heterogeneous scales. The simulator exposes three choices, selected by `angular_mode`:

<code>angular_mode</code>	λ	Comment
<code>field_norm</code> (default)	$ F_i $, capped at <code>lambda_max</code>	the hand-off report’s equation
<code>fixed</code>	<code>lambda_theta</code>	the $c_\bullet$ then set only a relative priority
<code>saturating</code>	$\lambda_{\max} F / (F + F_{\text{ref}})$	graded response, bounded by construction

The `fixed` mode is the clean extension of the Persistent Turning Walker and makes `lambda_theta` directly calibrable against the measured distribution of angular velocities; it is the recommended variant for the calibration phase. The default remains `field_norm` so that the simulator reproduces the reference model.

A strict priority logic was deliberately not adopted: it would introduce discontinuities in the drift field, breaking the Lipschitz-kernel assumption of the mean-field limit, and with it the bridge towards the macroscopic description.

2.3 Interaction kernels

Each term factorises into (strength) $\times$ (distance profile) $\times$ (angular profile) $\times$ (direction), with $\rho = |X_j - X_i|$, $\sigma_{ij} = (X_j - X_i)/\rho$, and the cone factor $A_\varepsilon(\alpha) = [(\varepsilon - (1 - \cos \alpha))/\varepsilon]_+$. Only two distance profiles are used: a *singular* one $(\ell - \rho)/\rho$ for the contact behaviours, and a *ramp* $(\rho - \ell_{\min})/(\ell_{\max} - \ell_{\min})$ for the attraction-at-distance behaviours.

Term	Groups	Support	Direction
braking ($\rightarrow \nu_b$)	all	$\rho \leq \ell_b$, frontal cone	— (radial)
avoidance	opposite	$\rho \leq \ell_{\text{avoid}}$, frontal cone	$\pm \sigma^\perp$ (disengaging)
antennation	opposite	$\ell_{\text{avoid}} < \rho \leq \ell_{\text{comm}}$, frontal cone	$+\sigma$
lateral repulsion	same	$\rho \leq \ell'_{\text{avoid}}$, lateral	$-\sigma$
overtaking	same	$\rho \leq \ell_{\text{over}}$, frontal cone, $u_j \leq \alpha \xi_i$	$\sigma^\perp$ towards goal
following	same	$\ell_f < \rho \leq \ell'_f$, frontal cone	$+\sigma$
objective	—	—	$k_{\text{obj}}(g, 0)$
wall	—	$\ell_{\min} < d \leq \ell_{\max}$	σ_{wall}

Every term can be switched on or off independently (`use_brake`, `use_comm`, ...), which makes ablations immediate.

2.4 Thigmotaxis: three laws

This is where the simulator departs most from the reference model, and the decision rests on a measurement. The hand-off report’s law (`wall_law = tangential`) weights the attraction by the *tangential* component of the heading: it is therefore *maximal when the agent is already walking along the wall*. Three measurements show that this makes the wall an absorbing state:

- an **isolated** agent, with no neighbour at all, starting from the centre, spends **98 %** of its time within 1 mm of a wall;
- the wall fraction **does not settle**: 24 % at 2 s, 38 % at 10 s, 40–46 % beyond 40 s (with no thigmotaxis: 3–6 %);
- it depends on bridge width at equal linear density: 47 / 36 / 23 % for $W = 5/10/20$ mm.

Mechanically: the force is *never* repulsive (zero if the agent is moving away, attractive otherwise), its radial profile *increases* with distance (zero at $\ell_{\min}$, maximal at $\ell_{\max} = 4$), and on a 10 mm bridge the active band covers 70 % of the width with its maximum immediately next to the centreline, of magnitude $10 = k_{\text{obj}}$. The centreline is therefore an unstable separatrix, and the wall band a well whose restoring force *grows* with the escape distance.

The simulator adds two laws. `outward_damping` only holds back an agent that is moving away; it fixes the weighting but remains one-sided and never initiates wall-following (it simply freezes y). `standoff` (default) restores the agent towards a target distance ℓ_{standoff} , **repulsive below it**, attractive above:

$$F_{\text{wall}} = c_{\text{wall}} w(d) \sigma_{\text{wall}}, \quad w(d) = \begin{cases} \frac{d-d^*}{\ell_{\max}-d^*} & d \geq d^*, \\ \frac{d-d^*}{d^*} & d < d^*. \end{cases}$$

What it produces, *measured* at the default parameters, is a wall **preference**, not an equilibrium: an isolated agent still wanders across the full width of the channel (distance to the wall from 0.5 to 5.0 mm, mean 1.9 on a half-width of 5), but it is never crushed against the edge (3 % of agents within 0.5 mm, against 30 % for `tangential`). That is the real gain of this law; the target distance is a soft attractor, not a fixed point.

Comparison on three criteria (single agent, head-on pair against the wall, crowd; `scripts/diagnose_wall.py`):

wall_law	single agent: final d	segregation	% crushed	speed
<code>tangential</code> , $c_{\text{wall}} = 10$	0.47 (crushed)	0.29	0.30	8.63
<code>outward_damping</code>	1.00, no capture	0.23	0.15	8.50
<code>standoff</code> , $c_{\text{wall}} = 10$	1.00	0.92 (lanes!)	0.12	13.42
<code>standoff</code> , $c_{\text{wall}} = 2$	1.00	0.32	0.11	9.61

$c_{\text{wall}} = 10$ had been calibrated for the **tangential** law, where the force is strongly attenuated; the **standoff** law applies the full c_{wall} to the offset from the target distance and therefore needs recalibrating. At $c_{\text{wall}} = 2$ all three criteria are met without degrading the mixing.

Known limitation, shared by all three laws: in an exactly symmetric configuration (two agents at the same ordinate, same tie-breaking sign) the avoidance steering cancels and the pair deadlocks. This is a measure-zero case — a 0.2 mm offset is enough for all three laws to cross cleanly — and it is what the report’s *wall-escape* term addressed; that term is kept as an option (**use_wall_escape**, disabled).

2.5 Boundary conditions

Side walls. **specular** (default) reverses the normal component of the heading, **slide** removes it. Both enforce zero normal flux; the report adopts specular reflection, which avoids the wall-sticking that a sliding condition induces at high density.

Bridge ends. **torus** (default): x is periodic and the ordinate is **redrawn at random on re-entry**. On a torus of finite length an agent passes the same place indefinitely; keeping its ordinate lets a transverse structure freeze in. **torus_keep_y** keeps y (continuous trajectory), **inflow** reinjects the departing agent at its group’s entrance. In all cases vision remains blocked at the ends (semi-periodicity).

The remixing has a measured effect on one observable: it destroys lanes by construction and therefore *underestimates* the effect of antenation on mixing. At $N = 150$ the segregation excess without antenation is $+0.25$ under **torus** against $+0.63$ under **torus_keep_y**. For that observable specifically, measurements should be made with **torus_keep_y**.

3 Parameters

Provenance: **M** = measured on the experimental trajectories, **C** = calibrated against a macroscopic observable, **I** = inherited from the reference model with no new constraint, **D** = modelling decision.

Group	Parameter	Default	Provenance
Geometry	length, height	100, 10	I (experimental bridge)
	radius	1.0	M – measured body 3.2×1.8 mm, equiv. radius
Perception	angle_vision, l_vision	π , 8	I
Radial	nu_cruise	20	M – see §3.2
	xi_cruise	14	M/C – compromise, see §3.1
	xi_spread	0.2	D – capped by decision
	nu_wall_stop	10^{10}	D – radial half of the no-flux condition
Angular	angular_mode	field_norm	I – see §2.2
	k_obj	10	I
	lambda_theta, lambda_max	10, 50	I
	sigma_theta	0.3	D – symmetry breaker, absent from the reference
Braking	l_brake	3.0	I – the data suggest ~ 5
	eps_brake, alpha_brake	0.7, 4	I
	c_brake	300	M – measured $\bar{\beta}$ 196 vs 175 in the model
Avoidance	l_avoid_steer, eps, c	2.3, 0.7, 50	I
Repulsion	l_repulse, eps, c	1.5, 2.0, 40	I
Antenation	l_comm_min, l_comm	2.3, 5.0	I
	eps_comm, c_comm	1.0, 12	C – 20 clogs over long runs
Overtaking	l, eps, c, alpha	4, 0.7, 15, 10	I
Following	l1, l2, eps, c	3, 8, 0.7, 30	I
Wall	wall_law	standoff	M – see §2.4
	c_wall	2.0	C – recalibrated for the standoff law

Group	Parameter	Default	Provenance
Boundaries	<code>l_wall_standoff, c_wall_damping</code>	1.0, 1.0	D – data suggest 2.5–4 mm
	<code>l_wall_min, l_wall_max</code>	0.5, 4.0	I
	<code>wall_mode, x_mode</code>	specular, torus	I / D
Simulation	<code>dt, record_fps</code>	0.01, 20	D

3.1 Cruising speed

Three observables give three incompatible values:

What one wants to match	required ξ
the slope of the diagram measured on the trajectories	~ 11 mm/s
the free-flow speed measured on the trajectories	12.5–13 mm/s
the published plateau $q_j = 10$	~ 15 mm/s

They cannot be satisfied together as long as ξ and $\bar{\beta}$ jointly set both the slope and the plateau — and the trajectory data **do not arbitrate**, since they stop at $k = 4.4$, well before the plateau. $\xi = 14$ is the compromise adopted: the plateau stays matched (10.20 against 10.0) and the discrepancy on the rise falls to +9–+30 %.

The best measurement of the free-flow speed is not “the speed of isolated ants” (rare at these densities, and the criterion also selects stopped ants) but the **plateau of speed as a function of the distance to the neighbour ahead**:

distance to the frontal neighbour (mm)	1.25	2.25	3.25	4.25	5.25	6.25	> 8
mean speed (mm/s)	5.35	7.83	11.30	12.85	13.20	12.71	~ 12.2

This profile also gives the braking range: the drop-off begins around **5 mm**, not 3. Lengths having been frozen by decision, the discrepancy is recorded but not corrected.

3.2 Speed relaxation rate

The correlation time of speed measured along the tracks is 0.227 s, which *seems* to impose $\nu_c \approx 4.4$. This is an over-interpretation: that time is not $1/\nu_c$, it is dominated by braking switching on and off as neighbours pass. Measured *with the same estimator on the simulation*:

	data	$\nu_c = 20$	$\nu_c = 4.4$
correlation time (s)	0.227	0.268	0.422

$\nu_c = 20$ is therefore the value *closest to the data*; lowering it moves away. Confirmed independently on the fundamental diagram: at constant ν_c/c_b ratio, varying ν_c by a factor 4.5 changes neither k_c ($17.6 \rightarrow 17.9$) nor q_j ($10.5 \rightarrow 10.1$). The diagram only sees the ratio; constraining ν_c will require a dynamical observable.

4 Numerical methods

Integration. Splitting scheme, with both relaxations solved *exactly* over the step at frozen field: $u \leftarrow u e^{-r\Delta t} + u^*(1 - e^{-r\Delta t})$ with $r = \nu_c + \nu_b$, and $\tan \frac{\theta - \omega}{2} \leftarrow \tan \frac{\theta - \omega}{2} e^{-\lambda \Delta t}$. Both are unconditionally stable and free of overshoot; $u \geq 0$ is preserved exactly (a convex combination of

positive quantities). Positions are then advected. The update is *synchronous*: everything is read from the current state and written to the next.

Neighbourhoods. Cell list of mesh ℓ_{vision} , amortised $O(N)$ cost, equivalent to the $O(N^2)$ sweep — verified neighbour by neighbour, sorted indices included. The walls of the rectangle are handled analytically (the reference code scanned 250 wall points per agent per step to find the nearest one, which is $\min(y, H - y)$), and visual occlusion by the wall is exact by convexity. No allocation inside the per-agent loop.

Parameters and compilation. No `@njit` function reads a parameter as a global variable: everything travels through a `NamedTuple` passed as an argument, and the discrete choices (geometry, angular mode, boundary condition) are integers branched inside the functions. Numba compiles once; any other instance then passes without recompilation, which is what makes sweeps possible.

Reproducibility. All randomness comes from a *single* `np.random.default_rng(seed)`. The reference code drew random numbers *inside* the compiled functions; but in nopython mode Numba holds its own random state, distinct from NumPy’s, which forced two generators to be seeded. The problem was removed rather than managed: degenerate direction tie-breaks use a sign drawn once per agent, and the noise and reinjection draws are pre-generated in blocks and passed to the kernel. A test automatically checks that no `np.random` call survives in compiled code. Draws are consumed at every substep whether or not the noise is active, so the random sequence does not depend on the model options: *ablations are comparable at equal seed*.

Storage. One run = one `config.json` (all parameters, the seed, the code version) and one `trajectories.npz` (compressed float32). The JSON alone suffices to replay the run identically, which `scripts/replay.py` verifies bit for bit.

5 Measurement methods

5.1 Fundamental diagram

Three flow estimators, measured on the *same* runs: (1) crossings of the central line with the density read in a central 1 cm^2 window — the eLife protocol, and the noisiest; (2) the same, averaged over 7 cross-sections; (3) whole domain, where k is exact and $q = \langle \sum |v_x| \rangle / (LW)$. In a stationary regime the mean flow is the same at every cross-section, so averaging cuts the variance without introducing bias. The three agree.

Caution with estimator (1) at high density: it measures a *local* density, which legitimately exceeds the global one when a cluster occupies the window (up to $k = 39$ for a global density of 28).

Two-phase fit. Phase 1: mono-kinetic closure $q = \xi \nu_c k / (\nu_c + \bar{\beta} k)$ on the low and mid densities. Phase 2: constant plateau on the high densities *only*. The split is chosen to maximise the overall R^2 , and k_c is the abscissa where the two branches meet.

Error bars. Standard deviation across seeds (the standard error is also in the CSV). They are small because each point is *already* a time average over ~ 24 s of stationary regime, i.e. ~ 480 frames: the instantaneous fluctuation is averaged out within a run, and all that remains between seeds is the variability of the initial condition.

5.2 Antennal contacts

A pair (i, j) is *in contact* at a given frame if: the groups are opposite; $\rho \leq d_{\text{contact}} = 2.5 \text{ mm}$ (the body is $2R = 2 \text{ mm}$, so the threshold places contact a quarter of a diameter beyond it, representing antennae touching); and **each** sees the other within a frontal cone of $\pm 60^\circ$ — the condition is mutual, so an ant being overtaken from behind is not an antennal contact.

An *episode* starts at the first frame where the condition becomes true and ends at the first frame where it becomes false. Resolution: 0.05 s; there is no hysteresis, so a pair flickering around

the threshold is counted several times — that is the main limitation.

C is the number of contacts experienced during a *transit* of a central 20 mm section: a continuous stay, with entry *and* exit observed, and no periodic wrap in between.

What depends on conventions. The *linearity* of C in k is robust (all conventions, all velocity parameter sets, $R^2 \geq 0.98$ up to $k \approx 20$): that is what is a property of the model. The *value* of the slope varies from 0.58 to 0.95 depending on the antennal range adopted (2.2 to 3.5 mm). Agreement at 0.61 therefore constrains the convention as much as it validates the model; a proper calibration would measure d_{contact} on the trajectories before comparing slopes.

Beyond $k = 20$ the fit stops: the flow has split into clusters, complete transits become long and rare, and the count becomes noisy (R^2 drops to 0.93).

5.3 Transverse mixing

The raw index is the occupancy-weighted mean of $|n_+ - n_-|/(n_+ + n_-)$ over 8 horizontal strips. **It is not zero for a perfectly mixed flow:** with n agents per strip and randomly assigned groups it averages $\sqrt{2/\pi n}$, i.e. 0.29 for 7 agents and 0.14 for 30. It therefore decreases mechanically as density rises, without any additional mixing taking place — a trap we fell into before correcting it.

The baseline is therefore *measured* rather than assumed: at each frame the group labels are reshuffled at random among the agents present (*positions are unchanged*) and the index is recomputed. This is the exact null “observed positions, groups independent of position”. Only the **excess** (raw minus baseline) is interpretable.

5.4 Space–time diagram

Local density per x -slice and per frame. At $k = 6$: diagonal streaks in both directions, no persistent structure. At $k = 25$: bright, persistent vertical bands — near-stationary clusters lasting tens of seconds alongside free-flowing regions — the phase separation that explains the plateau.

5.5 Comparison with the published measurements

Two points of comparison are available, and they must be distinguished.

The two-phase fit of the article (linear up to $k_j = 8$, then constant) is the one the hand-off report compared itself to. It is only one of four fits the article overlays on its data (two-phase, Underwood, Pipes–Munjál, Greenshields), and “ $k_j = 8$ ” is a parameter of it, not a measurement.

The published points (figure 4 of the article, panels A and B) are the measurements. They were read off by hand and stored in `donnees_publiees/elife2019_figure4.csv`; the accuracy is that of a graphical reading (~ 0.05 cm/s, ~ 0.2 ant/cm/s). These are what one should compare against, and they say two things the fit conceals:

- **the cloud has no knee:** $q(k)$ starts bending at $k \approx 3$ and saturates gradually towards $k \approx 12$ –14. The shape is that of a smooth saturating law, not a broken line — and that is precisely the shape the model produces;
- **speed decreases continuously** from $v \approx 17.5$ mm/s as $k \rightarrow 0$ down to ~ 7.5 mm/s at $k = 14$, with no intermediate plateau or bump.

5.6 Fundamental diagram of the data

Reconstructed from the raw trajectories with the same estimator as the simulations, on three recordings (colonies 800, 3200, 12800; bridges 10 and 20 mm; 25 frames/s, positions in mm). Each recording is split into 10 s windows.

A trap, now fixed: in the 12800 dataset the identifiers are **recycled**, so a label jumps from one end of the bridge to the other between two frames. These teleportations were being counted

as line crossings and produced absurd flows (q up to 66). We therefore require a displacement below 4 mm per frame, i.e. 100 mm/s, far above the real maximum speed ($p_{99} = 31$ mm/s).

The data cover $k \leq 4.4$: they constrain the rise, not the knee or the plateau.

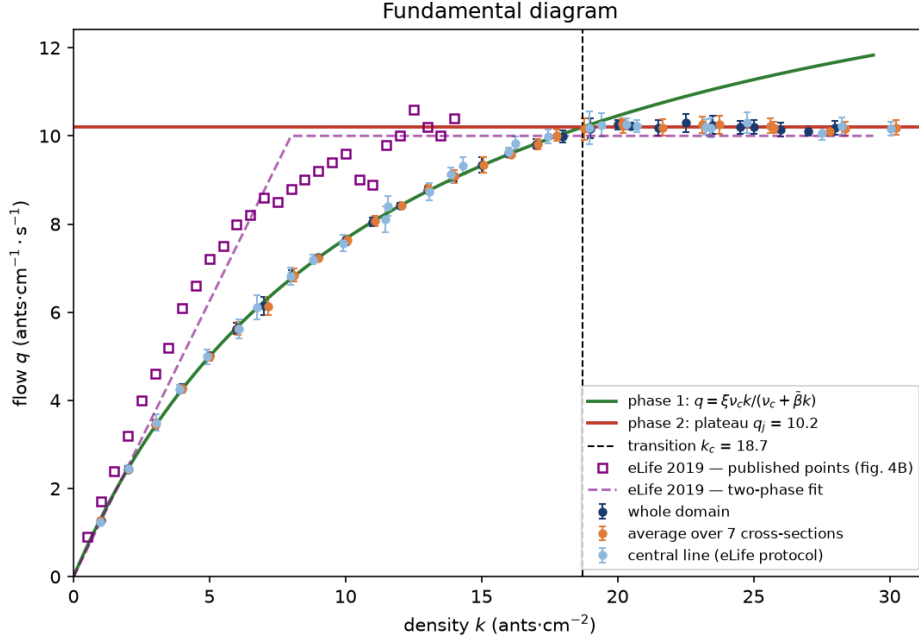


Figure 1: Fundamental diagram. Purple squares: published points of the article (fig. 4B); dashed line: their two-phase fit. The model reproduces the saturating shape and the plateau level, but reaches it later: between $k = 3$ and $k = 16$ it lies below the measurements.

6 Departures from the reference model

1. **Positional non-penetration removed** (user decision). The report enabled it ($g = 1.0$, $\kappa = 0.4$) and obtained $k_c \approx 12.4$, $q_j \approx 10.3$. Without it, overlap is no longer bounded, braking sees less physical density and saturates later. Measured cost: k_c moves from ~ 12 to ~ 19 .
2. **standoff wall law** instead of **tangential** (§2.4), with c_{wall} recalibrated from 10 to 2.
3. **Transverse remixing at periodic re-entry**, with its measured effect on the mixing observable.
4. **Deterministic tie-breaks** and a single random generator.
5. **ξ lowered from 17 to 14**, **xi_spread** kept at 0.2.

The report’s *wall-escape* term (`use_wall_escape`) is implemented but disabled; the `report` preset restores the report’s wall law and angular mode, but can no longer reproduce the published diagram exactly since non-penetration has been removed.

7 Reference results

Default configuration, 29 densities ($N = 10$ to 280), 6 seeds, 40 s, 174 runs.

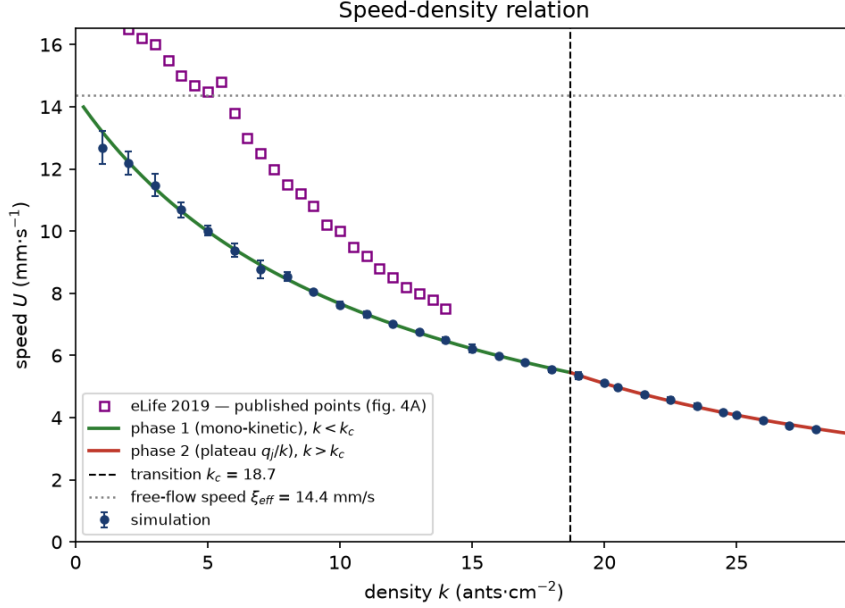


Figure 2: Speed–density relation. The offset is systematic: the published speed starts at 17.5 mm/s and decreases slowly at first, whereas the model starts at 14.4 and decreases fastest at low density. The two curves meet around $k \approx 16$.

Quantity	Simulation	Reference
Shape of the diagram	linear then strict plateau	two-phase
Phase-1 slope	14.4 mm/s	$v_f \approx 12\text{--}21$
Effective braking $\bar{\beta}$	174 mm ² /s	196 (measured on data)
Plateau q_j	10.21	≈ 10
Transition k_c	18.6	8 (fit) / 10–12 (cloud)
Contacts ($k \leq 20$)	$C = 0.67k + 0.54$	$C = 0.61k$
Time cost ΔT	0.15 s	0.24 s
Segregation excess	−0.03 to +0.09	—
without antennation	+0.21 to +0.23	—

Across the 11 densities between $k = 19$ and 28, the flow stays between 10.11 and 10.29: the plateau is *strictly flat*, with no decline whatsoever, up to one and a half times the maximum experimental density. The stopped fraction there rises from 3 to 12 %: the flow slows down a great deal but does not clog.

8 Limitations and open points

The knee does not move. None of the magnitudes or rates tested (k_{obj} , c_{avoid} , c_{repulse} , c_b , ν_c , xi_spread , σ_θ) brings k_c near 8: it stays between 13 and 19. The diagnosis lies in the speed *at* the transition, $U(k_c) = 10q_j/k_c$: the two-phase description requires 12.5 mm/s, i.e. the free-flow speed — hence no slowdown at all before the knee — whereas the model gives 4 to 6. Structurally, $q(8) = 10$ with $\xi = 12.5$ requires $\bar{\beta} \rightarrow 0$. Increasing c_b does bring k_c closer to 8 but lowers the plateau in the same proportion: the two are tied by the same constant.

But that knee does not exist in the data: the published points start bending at $k \approx 3$ and saturate gradually (§5.5). “ $k_j = 8$ ” is a parameter of the two-phase fit, not a measurement, and Underwood describes the cloud just as well. **The knee question is therefore largely a false problem:** the model has the right qualitative *shape*. What remains to be explained is not the

absence of a knee, but the fact that saturation arrives too late — a direct consequence of the next point.

Speed is too low at intermediate densities. The comparison with the published points (§5.5) is unambiguous:

k (ants/cm ²)	1	5	10	14
published U (mm/s)	17.2	14.5	10.0	7.5
model U (mm/s)	12.7	10.7	7.6	6.7

Two components, of different natures.

A scale offset. The published speed extrapolated to $k \rightarrow 0$ is ~ 17.5 mm/s, which supports $\xi \approx 17$ — the model’s original value, not the 12.5–13 given by the braking profile measured on the trajectories we hold. **The two data sources disagree by a factor ~ 1.4 on speed**, and this has not been resolved: the three available recordings (colonies 800, 3200, 12800) give a mean scalar speed of 10–12 mm/s where the published curve, which aggregates 170 experiments, gives 15–17. Before recalibrating ξ , the origin of that gap must be understood (trajectory smoothing protocol, inclusion or exclusion of stopped ants, different sets of experiments).

A shape mismatch. The published speed is nearly *flat* up to $k \approx 3$ and then decreases, i.e. sigmoidal; the Monod law, by contrast, decreases fastest at zero density. This means the real braking is **weaker than linear at low density and stronger at high density**. That is a structural constraint on the braking kernel which neither ξ nor $\bar{\beta}$ can satisfy, and it echoes the finding on the knee: it is not a tuning problem.

No stopped ants. The data contain **4.7 %** of speeds below 1 mm/s — the spike at zero in the histogram — against 0.1 % in the model at comparable density. This is the sharpest remaining gap, and precisely the mechanism on which Dobramysl et al. (2024) base their explanation of the plateau. First candidate for an extension.

Uncalibrated lengths. The measured braking range is ~ 5 mm against 3 in the model, and the wall standoff distance suggested by the transverse profiles is 2.5–4 mm against 1. Lengths were frozen by decision; they are the next batch to calibrate, and the speed profile as a function of distance to the frontal neighbour (§3.1) provides a direct method.

Antennation without an explicit time cost. The emergent ΔT (0.15 s) is of the right order but below the measured 0.24 s. A per-agent antennation timer remains the natural extension (item B.1 of the project backlog).

Not tested. Widths 5/10/20 mm, asymmetric foraging/returning composition, bottleneck geometries, priority rules.